

Databricks 14 Days AI Challenge

Day 1: Platform Setup & First Steps (09/01/26)

1. Why Databricks vs pandas/Hadoop?

Pandas is for single-machine analysis, Hadoop is for distributed storage and batch processing, and Databricks gives you distributed, in-memory analytics with ML and SQL in one platform. It combines the scale of Hadoop with the usability of Pandas.

2. Lakehouse architecture basics?

Lakehouse combines the low-cost storage of data lakes with the reliability and performance of data warehouses, using table formats like Delta Lake and engines like Spark to support BI, streaming, and ML on the same data.

3. Databricks workspace structure?

A Databricks workspace contains notebooks for development, clusters for compute, repos for version control, jobs for automation, and a shared Delta Lake for data, allowing analytics, engineering, and ML teams to work on the same platform.

4. Industry use cases (Netflix, Shell, Comcast)

Netflix uses Databricks for recommendations and churn prediction, Shell uses it for IoT analytics and predictive maintenance, and Comcast uses it for network optimization and customer analytics. In all cases, Databricks provides a single Lakehouse to run large-scale analytics and machine learning on the same data.